

ESP-WROOM-S2

Datasheet



Version 1.7
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About This Guide

This document introduces the specifications of ESP-WROOM-S2 hardware.

Release Notes

Date	Version	Release notes
2016.06	V1.0	First release.
2016.08	V1.1	- Updated the operating temperature range; - Added NCC Wi-Fi standard; - Updated Section 3.4 “Interface Description”.
2016.11	V1.2	- Added Appendix—Learning Resources; - Added “ESP-WROOM-S2 Peripheral Schematics” in Chapter 5.
2016.12	V1.3	- Changed the minimum working voltage from 3.0V to 2.5V; - Changed the power consumption during Deep-sleep from 10 μA to 20 μA.
2017.02	V1.4	Updated Section 3.3.
2017.09	V1.5	- Added Documentation Change Notification and the official link for downloading Product Certifications; - Updated the supply voltage to 2.7V ~ 3.6V; - Updated Figure 2-1 and added a note to it. - Updated Chapter 4: Combined electrical-characteristics-related data into Table 4-1; combined Wi-Fi-radio-related data into Table 4-2 and updated the output power parameters; updated Reflow Profile; - Updated Chapter 5 Schematics and added a note.
2017.10	V1.6	- Updated RF certification; - Update the chip output impedance to 39+j6Ω in Table 4-2; - Updated the note for the peripheral schematics.
2018.04	V1.7	- Updated the note for the peripheral schematics. - Updated the link of ESP8266 Apps and ESP8266 Hardware Resources in the appendix A.1.

Documentation Change Notification

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Certification

Download certificates for Espressif products from [here](#).

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1.

Overview

Espressif provides the SMD module—ESP-WROOM-S2 that integrates ESP8266EX. The module has been adjusted to achieve the best RF performance. We recommend using ESP-WROOM-S2 for tests or for further development.

Note:

For more information on ESP8266EX, please refer to [ESP8266EX Datasheet](#).

The module size is 16 ± 0.2 mm x 23 ± 0.2 mm x 3 ± 0.15 mm (see Figure 1-1). The flash used on this module is a 2-MB SPI flash connected to HSPI, with a package size of SOP 8-150 mil. The gain of the on-board PCB antenna is 2 dBi.

The ESP-WROOM-S2 works as the SDIO/SPI slave with the SPI speed of up to 8 Mbps.

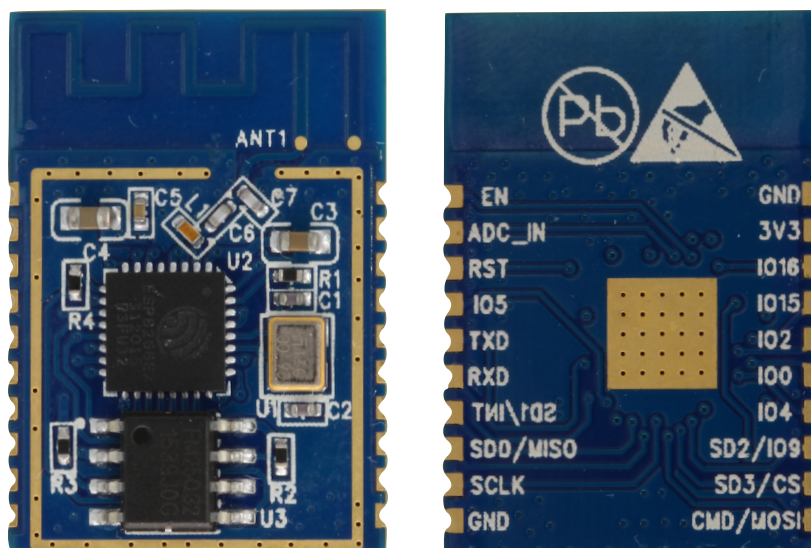


Figure 1-1. ESP-WROOM-S2 Module

Table 1-1. ESP-WROOM-S2 Specifications

Categories	Items	Specifications
Wi-Fi	RF Certification	FCC/CE
	Wi-Fi protocols	802.11 b/g/n
	Frequency range	2.4 GHz ~ 2.5 GHz (2400M ~ 2483.5M)
	Peripheral interface	UART/I2C/GPIO/PWM/SDIO/SPI/IR Remote Control/ADC
		GPIO/PWM



Categories	Items	Specifications
Hardware	Operating voltage	2.7V ~ 3.6V
	Operating current	Average: 80 mA
	Minimum current delivered by power supply	500 mA
	Operating temperature range	-40°C ~ 85°C
	Storage temperature range	-40°C ~ 85°C
	Package size (mm)	(16±0.2) x (23±0.2) x (3±0.15)
	External interface	-
Software	Wi-Fi mode	Station/SoftAP/SoftAP + Station
	Security	WPA/WPA2
	Encryption	WEP/TKIP/AES
	Firmware upgrade	UART Download / OTA (via network) / Download and write firmware via host
	Software development	Supports Cloud Server Development SDK for secondary development
	Network protocols	IPv4, TCP/UDP/HTTP/FTP
	User configuration	AT Instruction Set, Cloud Server, Android/iOS app

Note:

ESP-WROOM-S2 with high temperature range option (-40°C ~ 125°C) is available for custom order.



2.

Figure 2-1 shows the pin distribution of ESP-WROOM-S2.

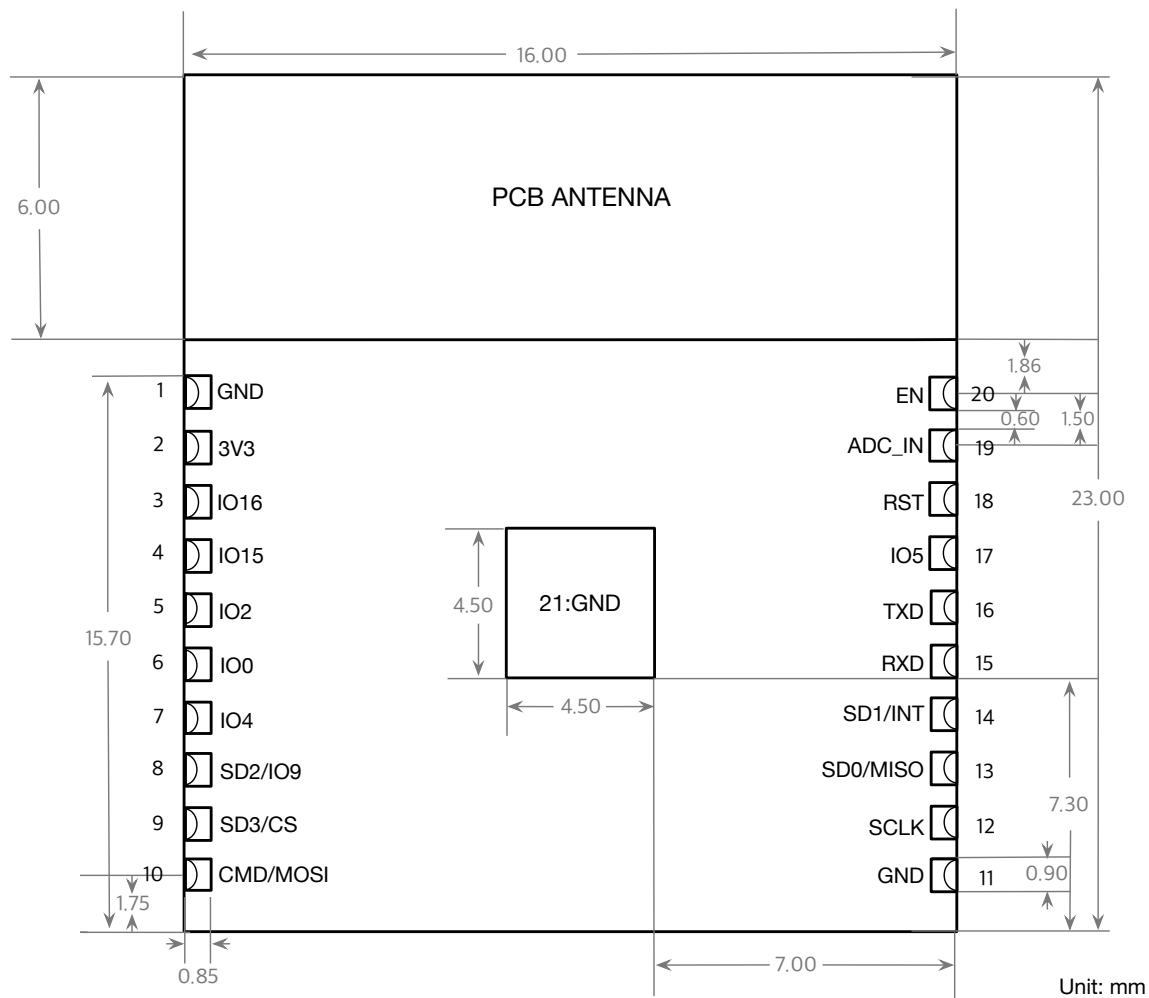


Figure 2-1. Top View of The ESP-WROOM-S2

Table 2-1. ESP-WROOM-S2 Dimensions (mm)

Length	Width	Height	PAD size (bottom)	Pin Pitch
16±0.2	23±0.2	3±0.15	0.9 x 0.85	1.5

ESP-WROOM-S2 has 20 pins, please see the pin definitions in Table 2-2.



Table 2-2. ESP-WROOM-S2 Pin Definitions

No.	Pin Name	Functional Description
1	GND	Ground
2	3V3	3.3V power supply (VDD)  Note: <i>It is recommended the maximum output current a power supply provides be of 500 mA or above.</i>
3	IO16	GPIO16; used for Deep-sleep wake-up when connected to RST pin.
4	IO15	HSPICS • UART download: pull down. • SDIO boot: floating (internal pull-up) or pull up.
5	IO2	GPIO2; UART1_TXD • UART download: pull down. • SDIO boot: don't-care.
6	IO0	GPIO0 • UART download: pull down. • SDIO boot: don't-care.
7	IO4	GPIO4
8	SD2/IO9	SD_D2 (Series resistor: 100 ~ 200Ω, 10k pull-up resistor); GPIO9
9	SD3/CS	SD_D3 (Series resistor: 100 ~ 200Ω, 10k pull-up resistor); SLAVE_SPI_CS
10	CMD/MOSI	SD_CMD (Series resistor: 100 ~ 200Ω, 10k pull-up resistor); SLAVE_SPI_MOSI
11	GND	Ground
12	SCLK	SD_CLK (Series resistor: 100 ~ 200Ω); SLAVE_SPI_CLK
13	SD0/MISO	SD_D0 (Series resistor: 100 ~ 200Ω, 10k pull-up resistor); SLAVE_SPI_MISO
14	SD1/INT	SD_D1 (Series resistor: 100 ~ 200Ω, 10k pull-up resistor); SLAVE_SPI_INT
15	RXD	UART0_RXD, receive end in UART download; GPIO3
16	TXD	UART0_TXD, transmit end in UART download, floating (internal pull-up) or pull up; GPIO1
17	IO5	GPIO5
18	RST	Reset
19	ADC_IN	Tests the power-supply voltage of VDD3P3 and the input power voltage of TOUT. These two functions cannot be used simultaneously.
20	EN	Chip enable pin (cannot be floating). Active high.



3. Functional Description

3.1. MCU

ESP8266EX contained in the ESP-WROOM-S2 integrates Tensilica L106 32-bit microcontroller (MCU) and a 16-bit RSIC. The CPU clock speed is 80 MHz and can reach a maximum value of 160 MHz. The system can readily run a Real-Time Operating System (RTOS). Currently, the Wi-Fi stack only takes up 20% of CPU time. The remaining CPU time (80% of total MIPS) can be used for user applications. The MCU can work in conjunction with the other parts of the chip through the following interfaces.

- Programmable RAM/ROM interface (iBus) that connects to the memory controller and can access the external flash.
- Data RAM interface (dBus) that connects to memory controller.
- AHB interface that accesses the register.

3.2. Memory

3.2.1. Internal SRAM and ROM

ESP8266EX Wi-Fi SoC integrates the memory controller including ROM and SRAM. MCU can access the memory controller through iBus, dBus, and AHB interfaces. All these interfaces can access ROM or RAM units. A memory arbiter determines the running sequence in the arrival order of requests.

According to our current version of SDK, SRAM space available to users is assigned as follows.

- RAM size < 50 kB, that is, when ESP8266EX is working in Station mode and connects to the router, available space in the Heap + Data sector is around 50 kB.
- There is no programmable ROM in ESP8266EX, therefore, user program must be stored in the SPI flash integrated into the ESP-WROOM-S2.

3.2.2. SPI Flash

ESP8266EX supports SPI flash. Theoretically speaking, ESP8266EX can support an up-to-16-MB SPI flash.

ESP-WROOM-S2 currently integrates a 2-MB SPI flash. ESP-WROOM-S2 supports these SPI modes: Standard SPI, DIO (Dual I/O), DOUT (Dual Output), QIO (Quad I/O) and QOUT (Quad Output).

**! Notice:**

Please use the most updated download tool and configure **SPI MODE** in the download tool as **DIO** or **DOUT**.

3.3. Crystal Oscillator

ESP-WROOM-S2 uses a 26 MHz crystal oscillator. The accuracy of the crystal oscillator should be ± 10 PPM.

When using the download tool, please note to select the right crystal oscillator type. In circuit design, capacitors C1 and C2 which connect to the earth are added to the input and output terminals of the crystal oscillator respectively. The values of the two capacitors can be flexible, ranging from 6 pF to 22 pF, however, the specific capacitive values depend on further testing of, and adjustment to, the overall performance of the whole circuit. Normally, the capacitive values of C1 and C2 are within 10 pF for the 26 MHz crystal oscillator.

3.4. Interface Description

Table 3-1. Interface Description

Interface	Pin	Functional Description
SPI	GPIO12/13/14/15 or GPIO6/7/8/11	S2 can control SPI Slave as a Master or communicate with Host MCU as a Slave. In overlap mode, S2 can share the SPI interface with Flash, shifted by different CS signals.
PWM	Any available GPIO (EXCEPT GPIO16)	Currently the demo provides four PWM channels (users can extend to six channels). PWM interface can realize the control of LED lights, buzzers, relays, electronic machines, etc.
IR	Any available GPIO (EXCEPT GPIO16)	The functionality of the infrared remote control interface can be realized via software programming. The interface uses NEC coding, modulation, and demodulation. The frequency of the modulated carrier signal is 38 kHz.
ADC	TOUT	Tests the power supply voltage of VDD3P3 (Pin3 and Pin4) and the input power voltage of TOUT (Pin6). However, these two functions cannot be used simultaneously. This interface is typically used in sensors.
I2C	Any available GPIO (EXCEPT GPIO16)	Connects to external sensors and display screens, etc.
UART	UART0: TXD(U0TXD), RXD(U0RXD) UART1: IO2(TXD)	Communicates with UART device. Downloading: U0TXD + U0RXD or GPIO2 + U0RXD Communicating (UART0): U0TXD, U0RXD Debugging: UART1_TXD (GPIO2) can be used to print debugging information.



4. Electrical Characteristics

Note:

Unless otherwise specified, measurements are based on $V_{DD} = 3.3V$, $T_A = 25^{\circ}C$.

4.1. Electrical Characteristics

Table 4-1. Electrical Characteristics

Parameter	Symbol	Min	Typ	Max	Unit
Storage temperature	-	-40	Normal	85	$^{\circ}C$
Operating temperature	-	-40	20	85	$^{\circ}C$
Maximum soldering temperature (Condition: IPC/JEDEC J-STD-020)	-	-	-	260	$^{\circ}C$
Supply voltage	VDD	2.7	3.3	3.6	V
Input logic level low	V _{IL}	-0.3	-	0.25 VDD	V
Input logic level high	V _{IH}	0.75 VDD	-	VDD + 0.3	V
Output logic level low	V _{OL}	-	-	0.1 VDD	V
Output logic level high	V _{OH}	0.8 VDD	-	N	V

4.2. Wi-Fi Radio

Table 4-2. Wi-Fi Radio Characteristics

Description	Min	Typ	Max	Unit
Input frequency	2400	-	2483.5	MHz
Input reflection	-	-	-10	dB
Output Power				
PA output power at 72.2 Mbps	13	14	15	dBm
PA output power in 11b mode	19.5	20	20.5	dBm
Sensitivity				
CCK, 1 Mbps	-	-98	-	dBm



Description	Min	Typ	Max	Unit
CCK, 11 Mbps	-	-91	-	dBm
6 Mbps (1/2 BPSK)	-	-93	-	dBm
54 Mbps (3/4 64-QAM)	-	-75	-	dBm
HT20, MCS7 (65 Mbps, 72.2 Mbps)	-	-72	-	dBm
Adjacent channel rejection				
OFDM, 6 Mbps	-	37	-	dB
OFDM, 54 Mbps	-	21	-	dB
HT20, MCS0	-	37	-	dB
HT20, MCS7	-	20	-	dB

4.3. Power Consumption

The following power consumption data were obtained from the tests with a 3.3V power supply and a voltage stabilizer, in 25°C ambient temperature. All data are based on 50% duty cycle in continuous transmission mode.

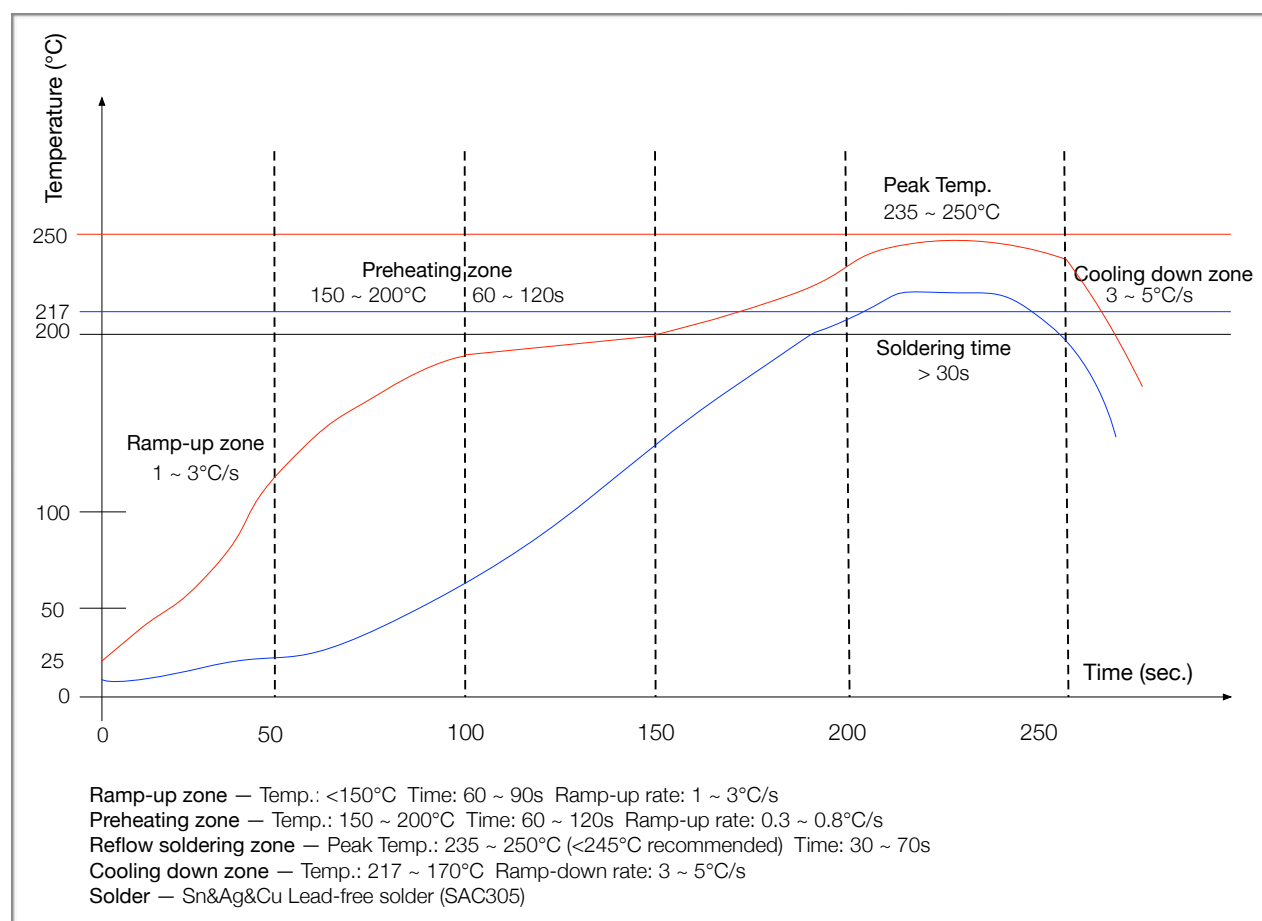
Table 4-3. Power Consumption

Modes	Min	Typ	Max	Unit
Tx 802.11b, CCK 11 Mbps, P _{OUT} = +17 dBm	-	170	-	mA
Tx 802.11g, OFDM 54 Mbps, P _{OUT} = +15 dBm	-	140	-	mA
Tx 802.11n, MCS7, P _{OUT} = +13 dBm	-	120	-	mA
Rx 802.11b, 1024 bytes packet length, -80 dBm	-	50	-	mA
Rx 802.11g, 1024 bytes packet length, -70 dBm	-	56	-	mA
Rx 802.11n, 1024 bytes packet length, -65 dBm	-	56	-	mA
Modem-sleep ^①	-	15	-	mA
Light-sleep ^②	-	0.9	-	mA
Deep-sleep ^③	-	20	-	μA
Power Off	-	0.5	-	μA

**Notes:**

- ① **Modem-sleep** is used when such applications as PWM or I2S require the CPU to be working. In cases where Wi-Fi connectivity is maintained and data transmission is not required, the Wi-Fi Modem circuit can be shut down to save power, according to 802.11 standards (such as U-APSD). For example, in DTIM3, when ESP8266EX sleeps for 300 ms and wakes up for 3 ms to receive Beacon packages from AP, the overall average current consumption is about 15 mA.
- ② **Light-sleep** is used for applications whose CPU may be suspended, such as Wi-Fi switch. In cases where Wi-Fi connectivity is maintained and data transmission is not required, Wi-Fi Modem circuit and CPU can be shut down to save power, according to 802.11 standards (such as U-APSD). For example, in DTIM3, when ESP8266EX sleeps for 300 ms and wakes up for 3 ms to receive Beacon packages from AP, the overall average current consumption is about 0.9 mA.
- ③ **Deep-sleep** is for applications that do not require Wi-Fi connectivity and only transmit data over long time lags, e.g., a temperature sensor that measures temperature every 100s. For example, when ESP8266EX sleeps for 300s then wakes up to connect to AP (taking about 0.3 ~ 1s), the overall average current consumption is far less than 1 mA. The current consumption of 20 μ A was obtained at the voltage of 2.5V.

4.4. Reflow Profile



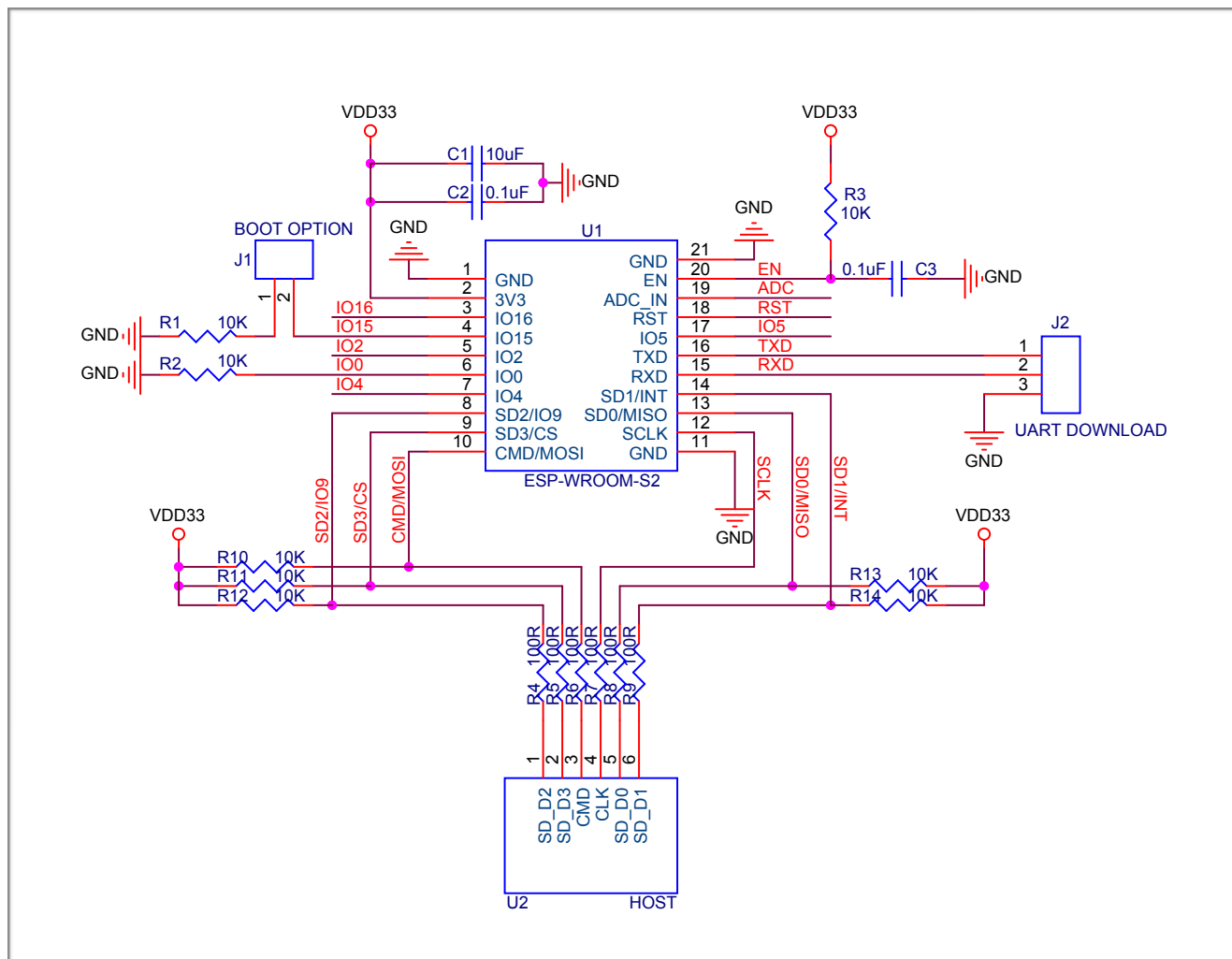


Schematics





6. Peripheral Schematics



Note:

Soldering Pad 21 to the Ground of the base board is not necessary for a satisfactory thermal performance. If users do want to solder it, they need to ensure correct quantity of solder paste.



A. Appendix—Learning Resources

A.1. Must-Read Documents

- [*ESP8266 Quick Start Guide*](#)

Description: This document is a quick user guide to getting started with ESP8266. It includes an introduction to the ESP-LAUNCHER, how to download firmware on to the board and run it, how to compile the AT application, structure and the debugging method of RTOS SDK. Basic documentation and other related resources for the ESP8266 are also provided.

- [*ESP8266 SDK Getting Started Guide*](#)

Description: This document takes ESP-LAUNCHER and ESP-WROOM-02 as examples to introduce how to use ESP8266 SDK. The contents include preparations before compilation, SDK compilation and firmware download.

- [*ESP-WROOM-02 PCB Design and Module Placement Guide*](#)

Description: The ESP-WROOM-02 module is designed to be soldered to a host PCB. This document compares six different placements of the antenna on a host board and provides notes on PCB design.

- [*ESP8266 Hardware Resources*](#)

Description: This zip package includes manufacturing specifications of the ESP8266 board and the modules, manufacturing BOM and schematics.

- [*ESP8266 AT Command Examples*](#)

Description: This document introduces some specific examples of using Espressif AT commands, including single connection as a TCP client, UDP transmission and transparent transmission, and multiple connection as a TCP server.

- [*ESP8266 AT Instruction Set*](#)

Description: This document provides lists of AT commands based on ESP8266_NONOS_SDK, including user-defined AT commands, basic AT commands, Wi-Fi AT commands and TCP/IP-related AT commands. It also introduces the downloading of AT firmware into flash.

- [*TCP/UDP UART Passthrough Test Demonstration*](#)

Description: This guide is intended to help users run a TCP & UDP passthrough test on the ESP8266 IoT platform.

- [*FAQ*](#)



A.2. Must-Have Resources

- [ESP8266 SDKs](#)

Description: This website page provides links to the latest version of ESP8266 SDK and the older ones.

- [ESP8266 Tools](#)

Description: This website page provides links to the ESP8266 flash download tools and ESP8266 performance evaluation tools.

- [ESP8266 Apps](#)

- [ESP8266 Certification and Test Guide](#)

- [ESP8266 BBS](#)

- [ESP8266 Resources](#)



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